

Activity 1: TL:DR

Prompt: Explain Google Transformers

The screenshot shows the ChatGPT web interface. At the top, there's a browser tab titled 'Axiom Login Testcases' and a URL bar with a long alphanumeric string. The ChatGPT header includes the logo, a 'Get Plus' button, and a 'Share' link. On the left, there's a sidebar with icons for chat, search, and a list of conversations. The main area displays a table of test cases:

2	AXIOM_INT_001	Verify successful EDM to Axiom interface data transmission with valid s
3	AXIOM_INT_002	Verify Axiom rejects EDM interface file with invalid file format
4	AXIOM_INT_003	Verify successful LDM to Axiom interface with valid transformed data
5	AXIOM_INT_004	Verify interface failure handling when EDM connection is unavailable
6	AXIOM_INT_005	Verify duplicate record handling during EDM interface processing
7	AXIOM_INT_006	Verify mandatory field validation during LDM interface processing
8	AXIOM_INT_007	Verify interface processing performance for high-volume EDM data load
9	AXIOM_INT_008	Verify encrypted file handling during EDM interface ingestion
10	AXIOM_INT_009	Verify interface audit logging for successful LDM data ingestion
11	AXIOM_INT_010	Verify corrupted file handling during LDM interface processing
12		
13		
14		

Below the table is a label 'Axiom_Interfaces_TestCases'. Underneath the table, there are icons for copy, like, dislike, and share. At the bottom, there's a text input field with the prompt 'Explain Google Transformers' and a microphone icon.

The screenshot shows the ChatGPT web interface with the response to the prompt 'Explain Google Transformers'. The response is as follows:

Google Transformers usually refers to the Transformer architecture introduced by [Google](#) researchers in the 2017 paper [Attention Is All You Need](#).

This architecture changed how AI models process language and became the foundation for modern LLMs like [OpenAI GPT](#) models, [Google Gemini](#), [Anthropic Claude](#), and many others.

Why Transformers Were Needed

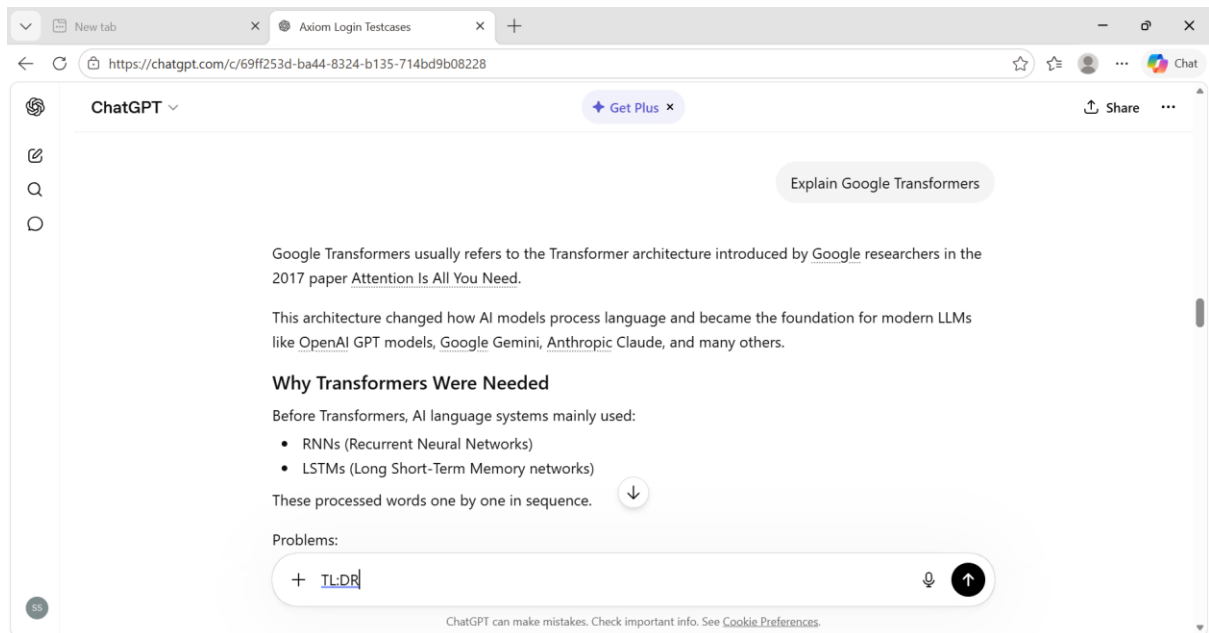
Before Transformers, AI language systems mainly used:

- RNNs (Recurrent Neural Networks)
- LSTMs (Long Short-Term Memory networks)

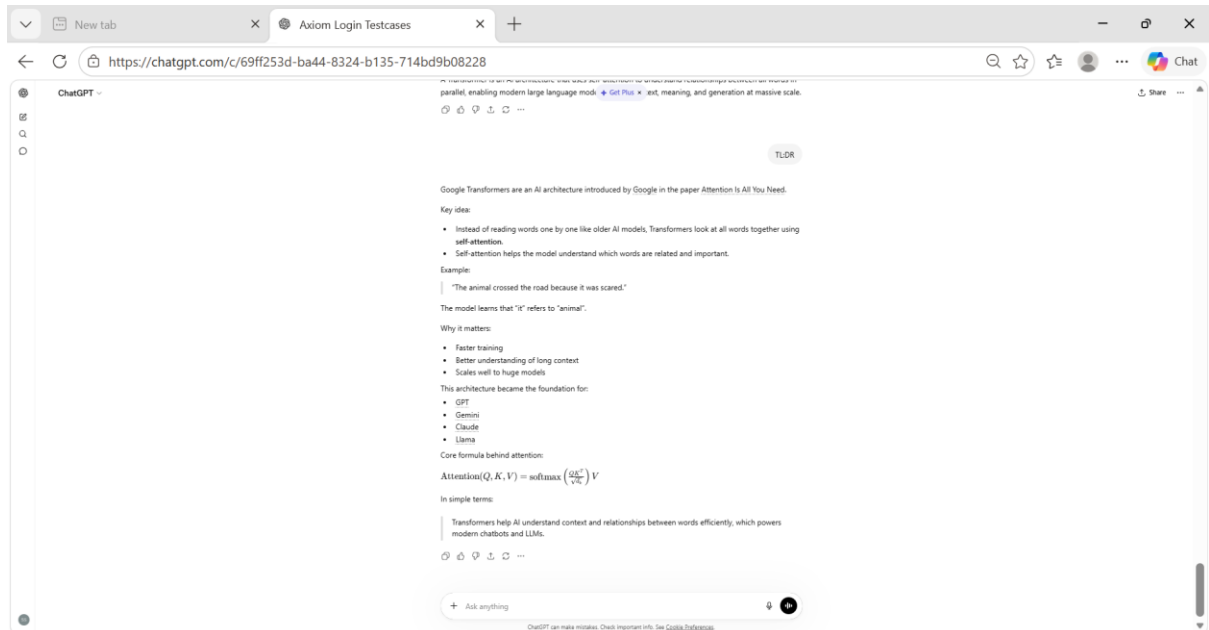
These processed words one by one in sequence.

Problems:

At the bottom, there's a text input field with the placeholder 'Ask anything' and a microphone icon.



TL:DR version:



Activity 2: ELI

Prompt: Explain Duckworth Lewis rule in cricket

The screenshot shows a ChatGPT chat window with the following content:

FFN(x) = Activation(xW₁ + b₁)W₂ + f + Get Plus

Where:

- x = input vector
- W_1, W_2 = learned weights
- Activation = ReLU or GELU

Real-World Analogy

Imagine:

- Attention = a group discussion
- Feed Forward = each person thinking privately after the discussion

The discussion gives context.
The private thinking improves understanding.

In Transformers

Each Transformer block contains:

1. Attention layer
2. Feed Forward layer

This pair repeats many times.

Example:

- Small models → few layers
- GPT-style LLMs → dozens or hundreds of layers

Important Insight

Attention helps the model decide:
| "What information matters?"

Feed Forward helps the model decide:
| "What should I do with that information?"

👍 🧐 📄 🔄 ...

+ Explain duckworth lewis rule in cricket

ChatGPT can make mistakes. Check important info. See [GPT-4o Disclaimers](#).

The screenshot shows a ChatGPT chat window with the following content:

Explain duckworth lewis rule in cricket

The Duckworth-Lewis-Stern (DLS) rule is a mathematical method used in limited-overs cricket to set a fair target when rain interrupts a match.

It is commonly used in:

- ODIs (50 overs)
- T20s

Developed originally by:

- Frank Duckworth
- Tony Lewis

Later updated by:

- Steven Stern

Core Idea

A batting team has two "resources":

1. Overs remaining
2. Wickets remaining

The more overs and wickets you have left:
→ the more runs you can potentially score.

Rain reduces overs.
So DLS recalculates targets based on remaining resources.

Simple Example

Normal Match

Team A:

- Scores 250 in 50 overs

Team B:

- Starts chase

Then rain happens.

Suppose Team B's innings is reduced from:

+ Ask anything

ChatGPT can make mistakes. Check important info. See [GPT-4o Disclaimers](#).

ELI 15 Version:

ChatGPT

Revised Target = Original Score × $\frac{\text{Rain}}{\text{Total Overs}}$ [Get Plus](#)

Actual DLS calculations are much more complex and use large statistical tables.

ELI15 Version

Think of cricket innings like fuel in a car.

Your "fuel" is:

- overs
- wickets

Rain removes some fuel.

DLS estimates:

"Given the fuel left, what is a fair target?"

Why DLS Sometimes Feels Weird

People often complain because:

- chasing teams sometimes get advantage
- interruptions change momentum
- T20 matches are highly unpredictable

But statistically, DLS is considered the fairest widely accepted system.

Famous Example

In the 2019 Cricket World Cup Final, rain interruptions and over calculations made DLS an important talking point during the tournament.

One-Line Summary

The Duckworth-Lewis-Stern rule recalculates cricket targets during rain interruptions by estimating how many scoring resources (overs + wickets) each team has remaining.

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Grokking Help](#)

Activity-3: Test Strategy generation using Chain of Thoughts (COT)

Prompt:

Context

Act as an experienced Test Manager in Finance and Risk module under Banking Domain

Instruction

Design a Test strategy document for Country Regulatory Reporting

[MANDATORY] Ask me 10 valid multiple-choice questions one by one

[STRICTLY] ONCE AFTER ANSWERING THE FIRST QUESTION Go with next question

[MANDATORY] Ask for approval after Senior Test Manager Review

[CRITICAL]Stick to the output format

Step by step

Example

Persona

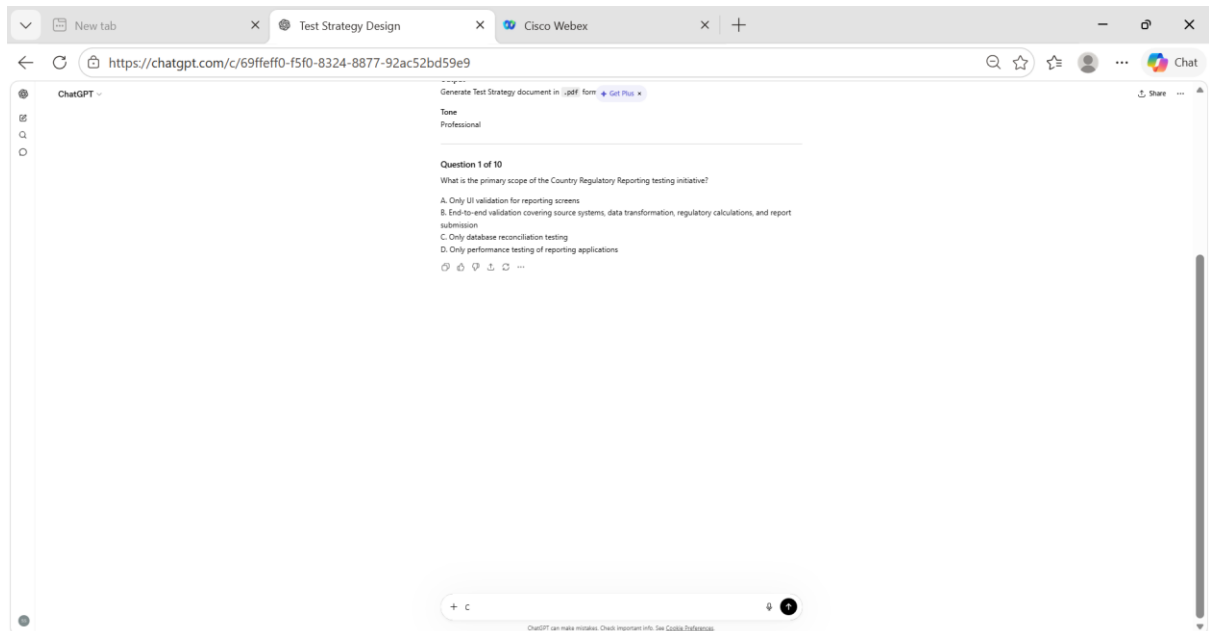
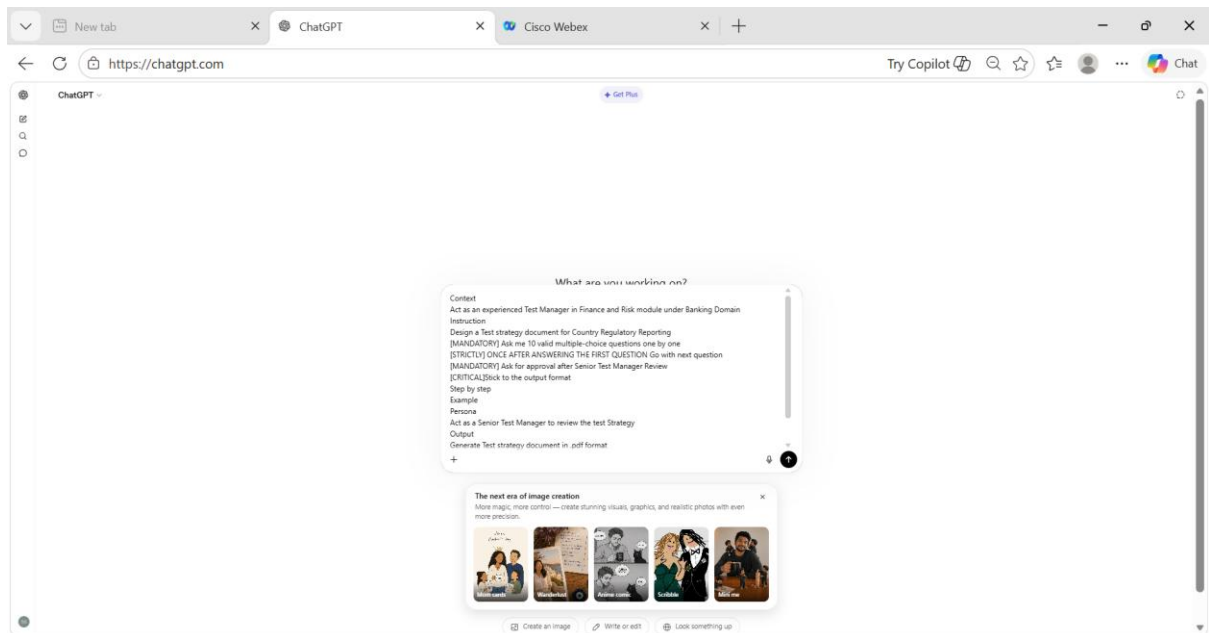
Act as a Senior Test Manager to review the test Strategy

Output

Generate Test strategy document in .pdf format

Tone

Generate the document in a professional tone



Browser tabs: New tab, Test Strategy Design, Cisco Webex

URL: <https://chatgpt.com/c/69ffeff0-f5f0-8324-8877-92ac52bd59e9>

ChatGPT

What is the primary scope of the Country Regulatory Testing initiative?

- A. Only UI validation for reporting screens
- B. End-to-end validation covering source systems, data transformation, regulatory calculations, and report submission
- C. Only database reconciliation testing
- D. Only performance testing of reporting applications

C

Persona
Act as an experienced Test Manager in Finance and Risk module under Banking Domain

Output
Generate Test Strategy document in .pdf format

Tone
Professional

Question 2 of 10
Which testing phase is considered most critical for validating regulatory reporting accuracy?

- A. Smoke Testing
- B. Unit Testing
- C. System Integration Testing (SIT)
- D. Browser Compatibility Testing

B and C

ChatGPT can make mistakes. Check important info. See [GPT-4o Limitations](#)

Browser tabs: New tab, Test Strategy Design, Cisco Webex

URL: <https://chatgpt.com/c/69ffeff0-f5f0-8324-8877-92ac52bd59e9>

ChatGPT

Question 2 of 10
Which testing phase is considered most critical for validating regulatory reporting accuracy?

- A. Smoke Testing
- B. Unit Testing
- C. System Integration Testing (SIT)
- D. Browser Compatibility Testing

B and C

Persona
Act as an experienced Test Manager in Finance and Risk module under Banking Domain

Output
Generate Test Strategy document in .pdf format

Tone
Professional

Question 3 of 10
Which data validation approach should be prioritized for Country Regulatory Reporting?

- A. Random sampling only
- B. Field-to-field reconciliation between source and target systems
- C. UI color and font validation
- D. Accessibility testing only

+

ChatGPT can make mistakes. Check important info. See [GPT-4o Limitations](#)

Browser tabs: New tab, Test Strategy Design, Cisco Webex

URL: <https://chatgpt.com/c/69ffeff0-f5f0-8324-8877-92ac52bd59e9>

ChatGPT

Question 10 of 10
Which reporting standard is commonly associated with Country Regulatory Reporting in banking?

- A. iFEG
- B. XML/XBRL
- C. MP3
- D. HTML only

Persona
Act as a Senior Test Manager to review the Test Strategy

Output
Generate Test Strategy document in .pdf format

Tone
Professional

Senior Test Manager Review Approval
Please confirm whether you approve proceeding with the final Country Regulatory Reporting Test Strategy Document generation based on your responses.

Options:

- A. Approved – Generate Final Test Strategy Document
- B. Approved with Minor Changes
- C. Not Approved – Rework Required

+ Approved

ChatGPT can make mistakes. Check important info. See [GPT-4o Data Sheet](#)

Browser tabs: New tab, Test Strategy Design, Cisco Webex

URL: <https://chatgpt.com/c/69ffeff0-f5f0-8324-8877-92ac52bd59e9>

ChatGPT

Please confirm whether you approve proceeding with the final Country Regulatory Reporting Test Strategy Document generation based on your responses.

Options:

- A. Approved – Generate Final Test Strategy Document
- B. Approved with Minor Changes
- C. Not Approved – Rework Required

Approved

Download the Country Regulatory Reporting Test Strategy PDF

You've reached your data analysis limit. Upgrade to ChatGPT Plus or try again after 545PM. [Upgrade](#)

+ Ask anything

ChatGPT can make mistakes. Check important info. See [GPT-4o Data Sheet](#)



Country_Regulatory_
Reporting_Test_Strate

Activity 4: Test Automation Framework using COT

Prompt:

Context

Act as an Automation Architect to design the framework for RestAssured API Automation with TestNG

Instructions

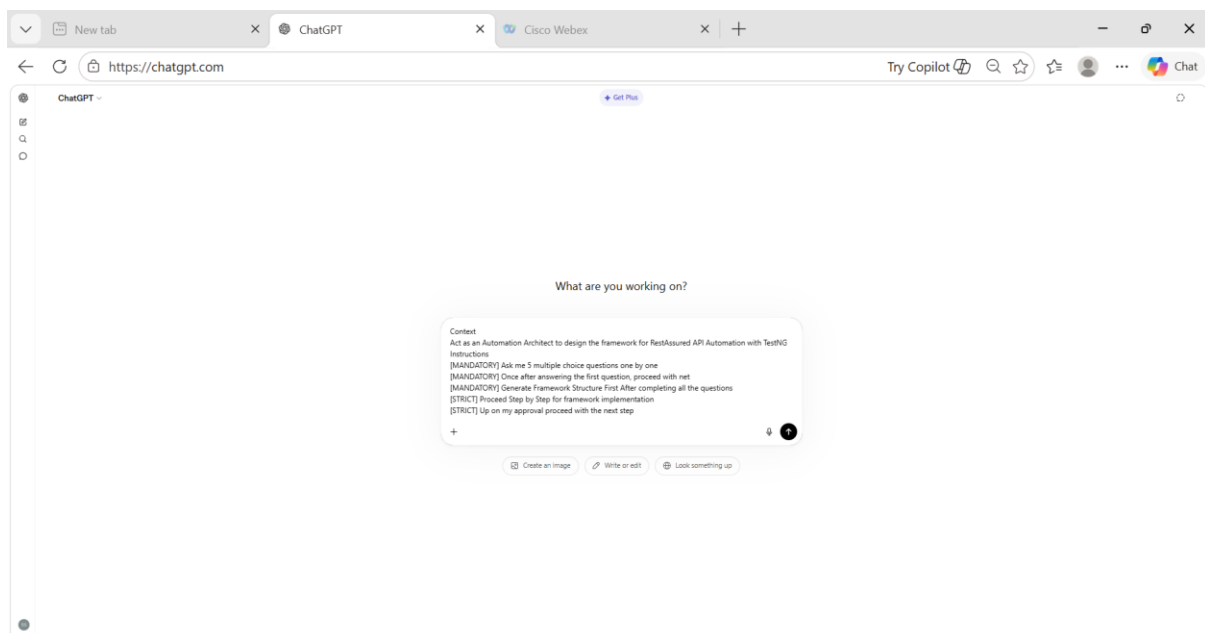
[MANDATORY] Ask me 5 multiple choice questions one by one

[MANDATORY] Once after answering the first question, proceed with net

[MANDATORY] Generate Framework Structure First After completing all the questions

[STRICT] Proceed Step by Step for framework implementation

[STRICT] Up on my approval proceed with the next step



New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

Context

Get Plus

Act as an Automation Architect to design the framework for RestAssured API Automation with TestNG

Instructions

(MANDATORY) Ask me 5 multiple choice questions one by one

(MANDATORY) Once after answering the first question, proceed with next

(MANDATORY) Generate Framework Structure First After completing all the questions

Show more

Question 1 of 5

What type of framework architecture do you prefer for the RestAssured + TestNG automation framework?

A. Layered Hybrid Framework (API Client + Utilities + Tests + Reporting)

B. BDD Framework using Cucumber with TestNG

C. Pure Data-Driven Framework using Excel/JSON

D. Microservice-wise Modular Framework with reusable common core

+ B

ChatGPT can make mistakes. Check important info. See [GPT-4o Data Sheet](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

Context

Get Plus

Act as an Automation Architect to design the framework for RestAssured API Automation with TestNG

Instructions

(MANDATORY) Ask me 5 multiple choice questions one by one

(MANDATORY) Once after answering the first question, proceed with next

(MANDATORY) Generate Framework Structure First After completing all the questions

Show more

Question 1 of 5

What type of framework architecture do you prefer for the RestAssured + TestNG automation framework?

A. Layered Hybrid Framework (API Client + Utilities + Tests + Reporting)

B. BDD Framework using Cucumber with TestNG

C. Pure Data-Driven Framework using Excel/JSON

D. Microservice-wise Modular Framework with reusable common core

+ B

ChatGPT can make mistakes. Check important info. See [GPT-4o Data Sheet](#)

Question 2 of 5

Which reporting mechanism should be integrated into the framework?

A. Extent Reports

B. Allure Reports

C. Both Extent and Allure

D. Default TestNG Reports only

+ A

ChatGPT can make mistakes. Check important info. See [GPT-4o Data Sheet](#)

Browser tabs: New tab, API Automation Framework, Qwen Studio, Online Shopping India Mobile, ChatGPT

URL: <https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb>

ChatGPT

Question 2 of 5
Which reporting mechanism should be integrated into the framework?

- A. Extent Reports
- B. Allure Reports
- C. Both Extent and Allure
- D. Default TestNG Reports only

Is this conversation helpful so far?

ChatGPT can make mistakes. Check important info. See [Grok's feedback](#)

Browser tabs: New tab, API Automation Framework, Qwen Studio, Online Shopping India Mobile, ChatGPT

URL: <https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb>

ChatGPT

Question 3 of 5
What should be the primary build and dependency management tool?

- A. Maven
- B. Gradle
- C. Ant
- D. No build tool required

Is this conversation helpful so far?

ChatGPT can make mistakes. Check important info. See [Grok's feedback](#)

Browser tabs: New tab, API Automation Framework, Qwen Studio, Online Shopping India Mobile, Ca...

URL: <https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb>

ChatGPT

Question 4 of 5
How should test data and environment configurations be managed?

- A. Properties files only
- B. Env-based configuration management
- C. JSON/YAML + Properties-based configuration
- D. Database-driven configuration management

Question 5 of 5
What level of CI/CD integration is expected for the framework?

- A. Jenkins integration only
- B. GitHub Actions only
- C. Jenkins + GitHub Actions + Parallel Execution support
- D. No CI/CD integration required

+ d

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

Browser tabs: New tab, API Automation Framework, Qwen Studio, Online Shopping India Mobile, Ca...

URL: <https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb>

ChatGPT

Proposed Framework Structure

```
RestAssuredBDDFramework/  
├── src/  
│   ├── test/  
│   │   ├── java/  
│   │   │   ├── api/  
│   │   │   │   ├── clients/  
│   │   │   │   │   ├── BaseAPIClient.java  
│   │   │   │   │   ├── endpoints/  
│   │   │   │   │   │   ├── UserEndpoints.java  
│   │   │   │   │   │   ├── payloads/  
│   │   │   │   │   │   │   ├── UserPayload.java  
│   │   │   │   │   │   ├── utilities/  
│   │   │   │   │   │   │   ├── ConfigReader.java  
│   │   │   │   │   │   │   ├── JsonUtils.java  
│   │   │   │   │   │   │   ├── VMLReader.java  
│   │   │   │   │   │   │   ├── LoggerUtility.java  
│   │   │   │   │   │   │   ├── ExtentManager.java  
│   │   │   │   │   │   ├── hooks/  
│   │   │   │   │   │   │   ├── Hooks.java  
│   │   │   │   │   │   ├── stepDefinitions/  
│   │   │   │   │   │   │   ├── UserStepDefinition.java  
│   │   │   │   │   │   ├── runners/  
│   │   │   │   │   │   │   ├── TestRunner.java  
│   │   │   │   │   │   ├── validations/  
│   │   │   │   │   │   │   ├── ResponseValidator.java
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Framework Implementation Plan (Step by Step)

Step 1 — Maven Project Setup

- Create Maven project
- Add dependencies:
 - RestAssured
 - TestNG
 - Cucumber
 - Extent Reports
 - Jackson/Gson
 - SLF4J/Log4j
- Configure Surefire plugin

Step 2 — Configuration Management

- Implement:
 - config.properties
 - environment.yaml
- Build ConfigReader utility

Step 3 — Base API Layer

- Create reusable RestAssured specifications
- Request/Response specifications
- Authentication handling
- Common headers management

Step 4 — BDD Layer Setup

- Configure Cucumber with TestNG
- Create Feature files
- Create Step Definitions
- Implement Hooks

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Step 5 — Payload & Test Data Management

- JSON payload builders
- Dynamic payload generation
- Externalized test data

Step 6 — Validation Layer

- Status code validations
- Schema validations
- Response body validations
- JSON path validations

Step 7 — Reporting & Logging

- Extent Report integration
- Failure screenshots/logging
- Request & response logging

Step 8 — Execution Strategy

- testng.xml configuration
- Tag-based execution
- Smoke/Regression suite setup

Step 9 — Framework Enhancements

- Retry mechanism
- Parallel execution readiness
- Token manager
- Environment switching

Step 1 Detailed Scope — Maven Project Setup

This step includes:

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Step 7 — Reporting & Logging

- Extent Report integration
- Failure screenshots/logging
- Request & response logging

Step 8 — Execution Strategy

- testng.xml configuration
- Tag-based execution
- Smoke/Regression suite setup

Step 9 — Framework Enhancements

- Retry mechanism
- Parallel execution readiness
- Token manager
- Environment switching

Step 1 Detailed Scope — Maven Project Setup

This step includes:

- Creating Maven project structure
- Creating pom.xml
- Adding required dependencies
- Configuring plugins
- Verifying project compilation

Please approve Step 1 to proceed with detailed implementation.

+ Approved

Lenovo Vantage

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Approved

Step 1 — Maven Project Setup Implementation

1. Create Maven Project

Project Name:

RestAssuredBDDFramework

Recommended Java Version:

Java 17

Recommended IDE:

IntelliJ IDEA

2. Standard Maven Folder Structure

```
RestAssuredBDDFramework
├── src
│   ├── test
│   │   ├── java
│   │   └── resources
│   └── main
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

2. Standard Maven Folder Structure

```
RestAssuredBDDFramework
├── src
│   ├── test
│   │   ├── java
│   │   └── resources
│   └── main
├── pom.xml
└── testing.xml
```

3. pom.xml Configuration

Edit

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.api.automation</groupId>
  <artifactId>RestAssuredBDDFramework</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Step 2 — Configuration Management Implementation

Objective

Implement centralized configuration management for:

- Environment handling
- Base URLs
- Authentication values
- Timeouts
- Common headers
- Execution configurations

This design ensures:

- Easy environment switching
- Zero hardcoding
- Better maintainability
- Reusability across APIs

1. Folder Structure for Configuration

```
src/test/resources/
├── config/
│   ├── config.properties
│   ├── environment.yaml
│   └── qa.properties
└── testData/
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

2. config.properties

Purpose:

- Global framework-level configurations
- Execution settings

Edit

Environment Selection
environment-QA

Request Timeout
connection.timeout=5000
read.timeout=10000

Reporting
extent.report.path=reports/ExtentReport.html

Logging
enable.request.logging=true
enable.response.logging=true

Authentication
auth.type=bearer

Default Content Type
content.type=application/json

3. environment.yaml

Purpose:

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

3. environment.yaml

Purpose:

- Store environment-specific details

Edit

QA:
baseUrl: "https://regres.in"
username: "qa_user"
password: "qa_password"
token: "qa_token"

UAT:
baseUrl: "https://uat.regres.in"
username: "uat_user"
password: "uat_password"
token: "uat_token"

PROD:
baseUrl: "https://prod.regres.in"
username: "prod_user"
password: "prod_password"
token: "prod_token"

4. ConfigReader Utility

Purpose:

- Read values from properties file
- Provide reusable getter methods

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

Browser tabs: New tab, API Automation Framework, Qwen Studio, Online Shopping India Mobile, Ca...

URL: <https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb>

ChatGPT

Edit

```
package api.utilities;

import java.io.FileInputStream;
import java.io.IOException;
import java.util.Properties;

public class ConfigReader {

    private static Properties properties;

    static {
        try {
            FileInputStream fis = new FileInputStream(
                "src/test/resources/config/config.properties");

            properties = new Properties();
            properties.load(fis);

        } catch (IOException e) {
            throw new RuntimeException("failed to load config.properties");
        }
    }

    public static String getProperty(String key) {
        return properties.getProperty(key);
    }
}
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookies](#) [Preferences](#)

Browser tabs: New tab, API Automation Framework, Qwen Studio, Online Shopping India Mobile, Ca...

URL: <https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb>

ChatGPT

Edit

6. Sample Usage

Fetch Base URL

```
String baseUrl =
    YAMLReader.getEnvironmentProperty("baseUrl");

System.out.println(baseUrl);
```

7. Recommended Best Practices

Do NOT hardcode:

- URLs
- Tokens
- Credentials
- Ports
- Headers

Always externalize:

- Environment configs
- Execution flags
- Runtime variables

Maintain separate:

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookies](#) [Preferences](#)

New tab

API Automation Framework

Qwen Studio

Online Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

- Reusable HTTP methods
- Centralized API execution layer

Please approve Step 3 to proceed.

Approved

Step 3 — Base API Layer Implementation

Objective

Create a reusable API engine layer that provides:

- Centralized RestAssured setup
- Reusable HTTP methods
- Request/Response specifications
- Authentication handling
- Common headers
- Logging support
- Better maintainability

This layer becomes the backbone of the framework.

1. Updated Framework Structure

```
api/
```

Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tab

API Automation Framework

Qwen Studio

Online Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

Approved

Step 4 — BDD Layer Setup Implementation

Objective

Implement the complete BDD automation layer using:

- Cucumber
- TestNG
- Feature files
- Step Definitions
- Hooks
- Runner classes

This layer enables:

- Business-readable test cases
- Reusable automation steps
- Better collaboration between QA and Business teams
- Scalable API test design

1. Updated Framework Structure

```
src/test/
├── java/
│   └── api/
│       └── hooks/
```

Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

Qwen Studio

Online Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

- Better collaboration between QA and business teams
- Scalable API test design

1. Updated Framework Structure

```
src/test/  
├── Save/  
│   └── api/  
│       ├── hooks/  
│       │   └── Hooks.java  
│       ├── runners/  
│       │   └── TestRunner.java  
│       ├── stepDefinitions/  
│       │   └── UserStepDefinition.java  
│       ├── validations/  
│       │   └── ResponseValidator.java  
└── resources/  
    ├── features/  
    └── User.feature
```

2. Feature File Creation

Purpose:

- Define business-readable API scenarios

Location:

src/test/resources/features/User.feature

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

Qwen Studio

Online Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

2. Feature File Creation

Purpose:

- Define business-readable API scenarios

Location:

src/test/resources/features/User.feature

Edit

Feature: User API Validation

@Smoke

Scenario: Validate Get User API

Given User API is available

When User sends GET request

Then Response status code should be 200

And Response should contain user data

3. Step Definition Implementation

Purpose:

- Connect Gherkin steps with Java code

Location:

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

ChatGPT

Please approve Step 5 to proceed.

Approved

Step 5 — Payload & Test Data Management Implementation

Objective

Implement a scalable payload and test data management layer for:

- Dynamic request body creation
- Externalized test data
- Reusable payload builders
- JSON serialization/deserialization
- Cleaner API automation code

This avoids:

- Hardcoded request bodies
- Duplicate payload logic
- Maintenance complexity

1. Updated Framework Structure

```
src/test/
├── ...
└── ...
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

ChatGPT

- request templates
- Environment-specific payloads
- Data providers with TestNG

11. Validation Checklist

Verify:

- Payloads serialize correctly
- JSON files load successfully
- POST APIs execute successfully
- Dynamic payloads work
- Response contains created data
- No hardcoded request bodies

Step 6 Preview — Validation Layer

Next step will implement:

- Response validations
- Status code validations
- JSON path validations
- Schema validations
- Header validations
- Assertion utilities
- Reusable validation engine

Please approve Step 6 to proceed.

Approved

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

Approved

Step 6 — Validation Layer Implementation

Objective

Implement a reusable and scalable response validation layer for:

- Status code validation
- Response body validation
- JSON path validation
- Header validation
- Schema validation
- Common assertions

This layer ensures:

- Centralized assertions
- Reduced duplication
- Cleaner step definitions
- Enterprise-grade validation handling

1. Updated Framework Structure

```
src/test/java/apl/|
|
| - validations/
|   | - ResponseValidator.java
|   | - SchemaValidator.java
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1ccea5cb

ChatGPT

Assertion utilitiesFaster development

JSON path validationsFine-grained checks

11. Validation Checklist

Verify:

- Status code validations work
- JSON path validations pass
- Headers validate correctly
- Schema validation succeeds
- Assertions fail properly for invalid responses
- Utilities are reusable

Step 7 Preview — Reporting & Logging

Next step will implement:

- Extent Report integration
- Log4j logging
- Request/response logging
- Failure screenshots/logging
- Scenario-level reporting
- Report lifecycle management

Please approve Step 7 to proceed.

+ Approved

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

Chat

ChatGPT

Approved

Share

Step 7 — Reporting & Logging Implementation

Objective

Implement enterprise-grade reporting and logging for:

- Extent Reports
- Scenario-level execution logs
- Request/Response logging
- Failure tracking
- Debugging support
- Execution visibility

This layer provides:

- Better traceability
- Faster issue analysis
- Audit-friendly execution reporting

1. Updated Framework Structure

```
src/test/java/api/
├──
├── utilities/
│   ├── ExtentManager.java
│   ├── LoggerUtility.java
│   └── ReportManager.java
├──
└── hooks/
```

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

New tabAPI Automation FrameworkQwen StudioOnline Shopping India Mobile, Ca

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

Chat

ChatGPT

Share

7. Request/Response Logging

Updated BaseAPIClient Example:

Edit

```
return RestAssured
    .given()
    .log().all()

    .spec(
        RequestSpecificationBuilder
            .getRequestSpecification()

        .when()
        .get(endpoint)

        .then()
        .log().all()

        .extract()
        .response());
```

8. Recommended Logging Strategy

Log Important Information

Log TypeRecommended

Request URLYes

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Suite segregation

Easier maintenance

13. Validation Checklist

Verify:

- Smoke suite executes correctly
- Regression suite executes correctly
- Tags filter scenarios properly
- Parallel execution works
- Reports generate correctly
- Maven commands execute successfully

Step 9 Preview — Framework Enhancements

Final step will implement:

- Retry mechanism
- Token manager
- Custom exception handling
- API chaining
- Reusable test context
- Parallel execution safety
- Enterprise-grade framework enhancements

Please approve Step 9 to proceed.

+ Approved

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)

API Automation Framework

https://chatgpt.com/c/69fff445-b420-8321-b1a5-1af1cceca5cb

ChatGPT

Verify end-to-end:

- APIs execute successfully
- Reports generate correctly
- Logs capture execution
- Payloads serialize properly
- Tags execute correctly
- Retry mechanism works
- Token handling works
- Framework supports scalability

Final Outcome

You now have a complete enterprise-grade:

RestAssured + TestNG + Cucumber BDD API Automation Framework

with:

- ✓ Scalable Architecture
- ✓ Reusable Components
- ✓ Centralized Configurations
- ✓ Dynamic Payloads Handling
- ✓ Validation Layer
- ✓ Reporting & Logging
- ✓ Execution Strategies
- ✓ Enterprise Enhancements
- ✓ Parallel Execution Readiness
- ✓ Maintainable Design

+ Ask anything

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)